

Experiment 8: Travelling Salesman Problem (TSP)

Aim

To implement the Travelling Salesman Problem using Artificial Intelligence techniques.

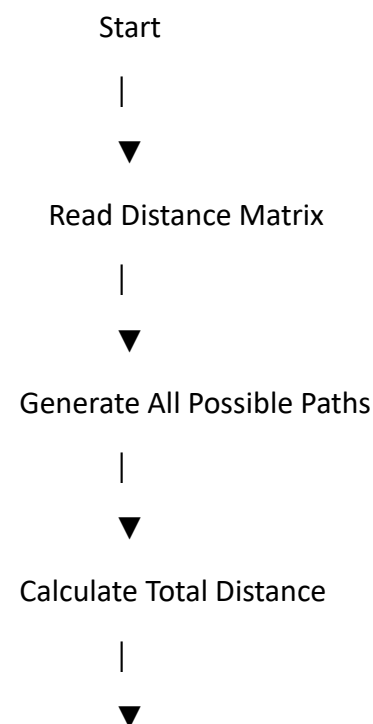
Objective

To find the shortest possible route that visits every city exactly once and returns to the starting city.

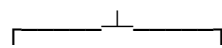
Algorithm

1. Define the distance matrix.
2. Generate all possible city permutations.
3. Calculate the total distance for each route.
4. Compare all route distances.
5. Select the route with the minimum distance.
6. Display the optimal path and its cost.

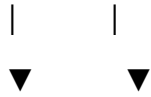
Flowchart



Is Distance Minimum?

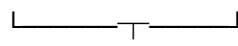


Yes No



Update Best Next Path

Path |



Display Result



End

Python Code

```
from itertools import permutations
```

```
graph = [  
    [0, 10, 15, 20],  
    [10, 0, 35, 25],  
    [15, 35, 0, 30],  
    [20, 25, 30, 0]  
]
```

```
cities = range(len(graph))
```

```
min_path = float('inf')
```

```
for path in permutations(cities):
```

```
    cost = 0
```

```
for i in range(len(path)-1):  
    cost += graph[path[i]][path[i+1]]  
  
cost += graph[path[-1]][path[0]]  
  
min_path = min(min_path, cost)  
  
print("Minimum Cost =", min_path)
```

Sample Output

Minimum Cost = 80

Result

Thus, the Travelling Salesman Problem was solved successfully and the minimum travel cost was obtained.